

Implementation and Results

Patient-Independent Localization of Myocardial Infarction from Vectorcardiographic Signals

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7 October 2026

The system we implemented

1. **Data and signal processing**

Audited PTB cohort, denoised VCG and aligned heartbeats.

2. **Paper reproduction**

Wavelet tensors, Tucker features and a random forest.

3. **Patient benchmark and representation study**

Nested model comparison and 12 feature ablations.

4. **External and acquisition-domain studies**

PTB-XL testing and paired measured/derived VCG analysis.

A Python command-line pipeline with saved configurations and results.

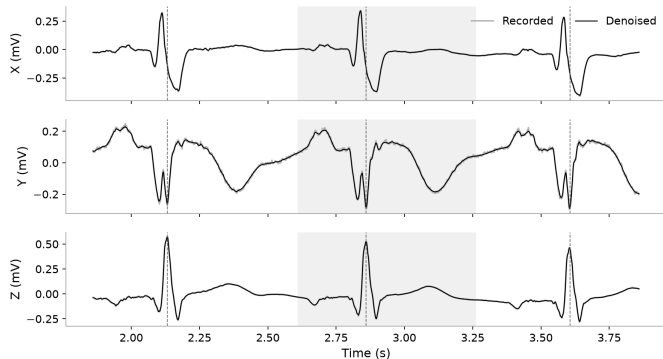
PTB data preparation

Cohort measure	Reproduced count
Independent patients	178
MI patients / healthy controls	126 / 52
VCG recordings	425
Complete heartbeats	60,507
Paper-reported heartbeats	60,527

- Measured Frank X, Y and Z signals at 1,000 Hz.
- Acute MI location labels and the paper cohort exclusions.
- Difference from the paper: 20 beats, or 0.033%.

Exact R-peak annotations and several preprocessing settings are unpublished.

Signal processing and beat extraction



Wavelet denoising
bior6.8 wavelet.

R-peak detection
Pan-Tompkins-style
detector on vector
magnitude.

Beat window
250 samples before R and
400 after R, including R.

651 samples per lead

Real PTB record patient001/s0010_re. Dashed lines: detected peaks. Shading: one beat.

Tensor construction and Tucker features

1. Retain the original beat and reconstruct details D3–D9.
2. Construct the lead $\times$ time $\times$ subband tensor:

$$\mathcal{X} \in \mathbb{R}^{3 \times 651 \times 8}.$$

3. Compress only time to rank 3 using truncated SVD:

$$\mathcal{G} \in \mathbb{R}^{3 \times 3 \times 8}, \quad \mathbf{f} = \text{vec}(\mathcal{G}) \in \mathbb{R}^{72}.$$

4. Classify with a 500-tree bootstrap random forest.

Lead and subband modes remain intact. SVD sign handling and flattening are deterministic.

The evaluation split changes the result

12-class evaluation	Beat accuracy
Published paper	99.80%
Our random beat 10-fold CV	99.136%
Our patient-grouped 10-fold CV	41.385%

- Random beat folds put beats from the same patient on both sides.
- Patient grouping keeps every patient's beats and recordings together.
- The same baseline loses **57.75 percentage points** when test patients are unseen.

Both reproduced scores measure beat accuracy. Patient overlap is zero in grouped CV.

Patient benchmark design

Class	Patients
AMI	17
ALMI	16
ASMI	27
IMI	29
ILMI	23
HC	52
Total	164

- Six classes with at least 10 patients each.
- 164 patients and 55,863 beats.
- 10 outer patient folds and 3 inner folds for tuning.
- Equal class weights and equal patient weights within each class.
- Average beat probabilities to make one patient prediction.

Primary endpoint: patient macro F1.

Every outer fold contains all six classes. Inner tuning uses only training patients.

Classifier comparison on Tucker features

Model	Patient accuracy	Patient macro F1
Random forest	47.56%	0.380
Linear SVM	43.90%	0.366
XGBoost	46.95%	0.403

- Same 72 features, patient folds and nested selection protocol.
- XGBoost leads on the prespecified macro F1 endpoint.
- The bootstrap intervals overlap substantially.

Six-class PTB task. The current cohort does not establish a definitive classifier winner.

Representation study with 12 ablations

Selected examples	Features	Patient macro F1
Wavelet statistics	192	0.464
Tucker rank 5	120	0.423
Tucker rank 3	72	0.416
Amplitude-normalized Tucker	72	0.369
No wavelet subbands	9	0.278

- Tested ranks, denoising, normalization, single leads and subbands.
- Wavelet statistics led the exploratory comparison.
- Same frozen folds and one fixed depth-6 XGBoost model.

Exploratory results differ from nested-selection estimates. All 12 results are in backup slide 16.

Wavelet statistics improved the PTB result

3 leads $\times$ 8 slices $\times$ 8 statistics = 192 features.

Mean, standard deviation, RMS, peak-to-peak amplitude, mean absolute value, log energy, skewness and kurtosis.

Nested PTB endpoint	Tucker + XGBoost	Statistics + XGBoost
Patient macro F1	0.403	0.487
Patient accuracy	46.95%	54.88%

- Macro F1 gain: **+0.084**; paired 95% interval: +0.003 to +0.165.
- Representation selection used these same PTB folds.

A nested rerun does not remove representation-selection bias. Fresh confirmation is required.

Locked external validation on PTB-XL

- Train on 164 eligible PTB patients and test once on PTB-XL fold 10.
- Strict test cohort: **1,185 records from 1,092 patients**.
- Convert 12-lead ECG to Kors-derived XYZ and resample 500 to 1,000 Hz.
- No PTB-XL model, threshold or representation tuning.

External patient endpoint	Result
Model macro F1	0.242
Model accuracy	56.23%
Always-healthy accuracy	71.98%

Healthy controls are 72.0% of this cohort. Localization does not transfer reliably.

Paired measured and derived VCG study

Same 164 PTB patients, 391 records and 55,863 original beat positions.

Training VCG	Test VCG	Patient macro F1
Measured Frank	Measured Frank	0.469
Measured Frank	Kors-derived	0.422
Kors-derived	Kors-derived	0.467

- Estimated measured-to-derived change: -0.046 **macro F1**.
- Paired 95% interval: -0.108 to $+0.017$, including zero.
- Conversion alone does not explain the external performance gap.

Fixed depth-3 XGBoost. Use the internal 0.469 control, not the earlier nested 0.487 estimate.

Codebase and reproducible experiments

Python package

- Data preparation and signal processing.
- Feature extraction and evaluation.
- Benchmarks and ablations.
- External and paired domain studies.

Repeatable experiments

- Saved YAML settings and patient folds.
- Cached feature banks.
- Predictions, metrics and confusion matrices.
- **15 automated tests passing.**

```
mi-localization run --mode both
```

```
mi-localization benchmark --models rf svm xgboost
```

Repository: https://github.com/Kumaryan12/major_project. Large signals and model caches stay local.

Completed work and next milestone

Completed

- Paper pipeline and close reproduction.
- Evaluation on unseen patients.
- Classifier and representation studies.
- External and paired domain experiments.

Proposed next milestone

- Compact 1D CNN or ResNet.
- Tensor or wavelet features combined with learned features.
- Same patient endpoint and evaluation folds.
- Fresh independent confirmation.

The main contribution so far is a tested research baseline.

Current evidence does not establish state-of-the-art performance or clinical suitability.

Backup: class-level results

Class	PTB patients	Tucker F1	Statistics F1	External F1
AMI	17	0.188	0.357	0.049
ALMI	16	0.400	0.471	0.000
ASMI	27	0.412	0.373	0.380
IMI	29	0.412	0.438	0.241
ILMI	23	0.263	0.450	0.035
HC	52	0.742	0.835	0.747

- AMI and ILMI improve on PTB; ASMI declines.
- PTB and PTB-XL use different cohorts and label mappings.
- External ALMI has only five patients and zero correct classifications.

Patient-level class F1 from the nested PTB comparisons and fixed external test.

Backup: full exploratory ablation results

Representation	Features	Patient accuracy	Patient macro F1
Wavelet statistics	192	53.66%	0.464
Tucker rank 5	120	48.17%	0.423
Tucker rank 8	192	49.39%	0.416
Tucker rank 3	72	48.17%	0.416
Raw signal, rank 3	72	47.56%	0.411
Tucker rank 2	48	45.12%	0.377
Amplitude normalized	72	43.29%	0.369
Z lead only	24	42.68%	0.363
Tucker rank 1	24	44.51%	0.353
X lead only	24	37.80%	0.320
Y lead only	24	35.37%	0.301
No wavelet subbands	9	35.37%	0.278

Same frozen PTB folds and fixed depth-6 XGBoost. These are exploratory scores.

Backup: technical choices and reproduction limits

Choice	Implementation and limit
Wavelet boundaries	Symmetric extension and soft thresholding.
R-peak detection	Vector magnitude; threshold 0.06 calibrated to beat counts.
Feature determinism	SVD sign fixing and column-major core flattening.
Classifier analogue	scikit-learn bootstrap forest instead of MATLAB TreeBagger.
Exact replication	Unpublished annotations, folds and options limit exact matching.

Reference

Zhang et al., *Automated Localization of Myocardial Infarction From Vectorcardiographic via Tensor Decomposition*, IEEE TBME, 70(3), 812–823, 2023. DOI: 10.1109/TBME.2022.3202962.

The code documents every implementation choice. Matching total beats does not match every annotation.